

Five-Plot Insight Report

Dataset: workplace_eda_training_dataset.csv (720 learners, 16 columns) | **Target:** completed_training (base rate 87.5%)

Data story

The headline question is why some learners finish workplace training and others do not. The five-plot exploration points to a clear, actionable answer: **engagement and friction matter; delivery format and prior skill barely do.**

The **assessment-score distribution** shows completers concentrated between 75 and 95 with a heavy peak at 100, while the smaller non-completer group sits lower and broader, centred around 70–80. The two groups overlap but separate visibly — assessment_score discriminates, though the chart raises two governance flags. First, the ceiling spike at 100 is unusual: a clean assessment process produces a smoother right tail, so the spike hints at either a scoring cap or a post-training pass-mark adjustment. Second, only completers reach the top of the scale, which suggests assessment_score may be measured *after* the completion decision, making it a leakage candidate rather than a usable input feature.

The **completion-rate-by-training-mode** bar chart is, deliberately, a flat one. Self-paced (88.2%), blended (87.2%) and instructor-led (86.9%) deliver almost identical completion rates. This is a useful negative result: the business should not expect to lift completion by changing delivery format alone.

The **correlation heatmap** ranks the candidate features. The strongest correlation with the target is productivity_gain_estimate_pct ($r = 0.50$), followed by engagement_score (0.27), satisfaction_score (0.22), and manager_support_score (0.20). Prior Python score and total course minutes show essentially no signal.

The **engagement-vs-completion** boxplot makes the strongest case for a single pre-completion feature. The completer median is ~80, the non-completer median ~69 — a 10-point gap with limited overlap in the inter-quartile boxes.

The **stakeholder chart** translates this for a learning manager: completion ranges from 82% in Customer Support to 93% in IT. The 11-point gap is large enough to justify targeted intervention rather than a uniform rollout.

Bottom line for modelling: engagement and manager-support are the most defensible features; productivity-gain looks tempting but is almost certainly outcome-contaminated.

Word count: 314

Hypotheses for the next modelling sprint

- **H1 — Engagement is predictive.** engagement_score shows clear separation between completers and non-completers; expected to be a top feature.
- **H2 — Manager support is predictive.** manager_support_score correlates 0.20 with completion and is available before training begins.
- **H3 — High friction signals risk.** A derived high_friction_flag ($\text{support_tickets_last_30d} \geq 3$) captures a non-linear drop from ~89% to 78% completion.
- **Leakage caveat** — productivity_gain_estimate_pct, assessment_score, and satisfaction_score all correlate strongly with the target but are measured at or after course end. Excluded from any pre-completion prediction model.
- **Sample-size caveat** — only 91 learners sit in the 3+ ticket band and just 5 have 5 or more tickets; tail-band estimates are noisy and should be reported with confidence intervals.